

Bhavana Sudhakar Athavane

Fullerton, CA | C: (657) 751-9230 | E: bhavana.sa@csu.fullerton.edu | [LinkedIn](#) | [GitHub](#)

Education

California State University, Fullerton

Master of Science, Computer Science

Fullerton, CA

May 2026

Jyothy Institute of Technology

Bachelor of Engineering, Computer Science

Bangalore, India

July 2022

Experience

California State University

Graduate Teaching Assistant – ISDS 361B (Business Analytics II)

Fullerton, CA

January 2026 – Present

- Mentored 50+ students per semester in applied business analytics, achieving a 95% assignment completion rate and improving average quiz scores by 18% over course duration through targeted feedback and lab sessions
- Delivered hands-on instruction in advanced Excel (VLOOKUP, INDEX-MATCH, Pivot Tables, Power Query, Macros) and SQL for structured data analysis across 10+ real-world datasets exceeding 100K rows

Nokia

Database Software Engineer – Co-Op

Sunnyvale, CA

September 2025 – December 2025

- Designed and optimized relational database schemas across Oracle, MySQL, and PostgreSQL for AI-driven backend systems, reducing average query latency by 55%
- Engineered complex SQL queries, stored procedures, and multi-level indexing strategies that cut ETL pipeline runtime by 65%
- Built Python-based REST API integrations connecting backend services to OLTP databases, reducing end-to-end system latency

Platform Engineering Architect – Co-Op

May 2025 – August 2025

- Engineered an AI-powered fault diagnostic pipeline on Google Cloud Vertex AI processing network telemetry data, achieving 82% fault classification accuracy and reducing Mean Time to Resolution (MTTR) by 41%
- Architected scalable log-processing pipelines on GCP (Pub/Sub, Dataflow, BigQuery) handling real-time telemetry data, cutting manual triage effort by 80% and enabling real-time alerting across 200+ network nodes
- Integrated LLM-powered semantic search (RAG architecture) across CMDB, JIRA, and distributed knowledge bases, surfacing relevant runbooks 3× faster and reducing incident resolution time by over 35%

Cognizant

Senior Systems Engineer

Bangalore, India

December 2023 – August 2024

- Led Oracle DB performance engineering initiative improving overall system throughput by 30% and supporting SLA compliance
- Developed automated SQL data-validation framework using Python and PL/SQL across Oracle DB and SQL Server, reducing reporting inconsistencies by 20%

Programmer Analyst Trainee

March 2022 – December 2023

- Automated 8 internal ETL workflows using Python and SQL, improving end-to-end process efficiency by 20%, enabling the team to reallocate 15 hours/week to higher-value analytics work
- Conducted multi-source data validation across Oracle DB and SQL Server for production pipelines; delivered 6 executive Tableau dashboards tracking KPIs across revenue, operations, and compliance

Technical Skills

- **Programming:** Python, SQL, Java, JavaScript, ETL Pipelines, Database Design, Query Optimization, Stored Procedures
- **Cloud & AI:** Google Cloud Platform (GCP), Vertex AI, BigQuery ML, Cloud Storage, LLM Integration, Semantic Search, NLP
- **Database:** Oracle DB, MySQL, SQL Server, AWS Redshift, PostgreSQL
- **Tools:** Git, Tableau, Jira REST APIs Redshift, PostgreSQL

Projects

AI-Powered Financial Fraud Detection System

- Engineered an end-to-end fraud detection pipeline integrating large-scale transaction ingestion, feature engineering, and AI-based anomaly classification using Vertex AI to identify high-risk financial patterns

Content Modelling using NLP

- Developed an NLP-driven document processing pipeline leveraging CNN models and OCR to extract, structure, and summarize unstructured textual data for downstream analytics

Textile Database Management System

- Designed and implemented a relational database system supporting real-time inventory, order processing, and payment workflows with normalized schema design and query optimization

E-Commerce Web Application (Internship Project)

- Built a React-based Amazon clone implementing authentication, cart persistence, and dynamic product browsing